

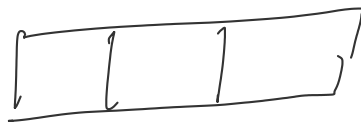
# Backtracking

Saturday, 14 February 2026

7:14 PM

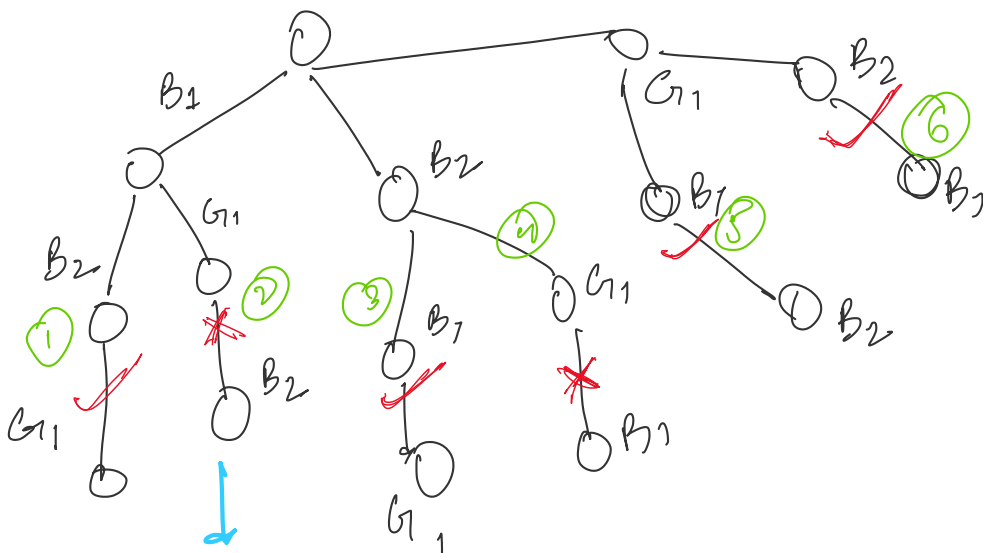
Brute force approach : (DFS) - depth search

→ This is used when we have multiple solutions and we want all these solutions. This will have some constraints & we will try to check these constraints & get soln which satisfy these constraints.  
Ex. If we have 2 boy and 1 girl and have 3 chairs then in how many ways can we arrange?

 $B_1 B_2 G_1$ 
$$n=3 \Rightarrow 3!$$

contraint  
↓  
Gist should be  
in middle.

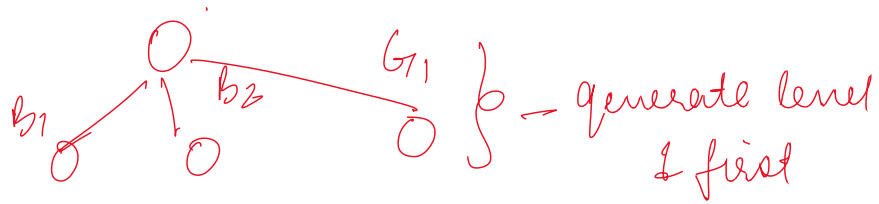
State space Tree  $\rightarrow$



bounding  
function.

## # Branch and Bound Strategy :: (BFS)

↳ This also follows brute force approach & generate space tree -



So we are generating here breadth first -